Department of Mathematics, IIT Madras
MA2101 Linear Algebra via Matrices
Assignment 5

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1. Suppose $A \in \mathbb{F}^{n\times n}$, $\lambda \in \mathbb{F}$, and $b \in \mathbb{F}^{n\times 1}$ are such that $(A - \lambda I)x = b$ has a unique solution. Show that λ is not an eigenvalue of A.
2. Formulate a converse of the statement in Exercise 1 and prove it.
3. Let $A \in \mathbb{F}^{n\times n}$. Show that a scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A iff the map $A - \lambda I : \mathbb{F}^{n\times 1} \to \mathbb{F}^{n\times 1}$ is not one-one.
4. Let $A \in \mathbb{F}^{n\times n}$. Show that A is invertible iff 0 is not an eigenvalue of A.
5. Let $A \in \mathbb{F}^{n\times n}$. If the entries in each column of A add up to a scalar λ, then find an eigenvector of A.
6. Find all eigenvalues and corresponding eigenvectors of the following matrices:

   (a) $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$
7. Find all eigenvalues and their corresponding eigenvectors of the $n \times n$ matrix A whose jth row has each entry j.
8. Let $p(t) = t^3 - \alpha t^2 - (\alpha + 3)t - 1$ for a real number α. Consider

$$A = \begin{bmatrix} \alpha & \alpha+3 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad P = \begin{bmatrix} -1 & 2 & -\alpha-3 \\ 1 & -1 & \alpha+2 \\ -1 & 1 & -\alpha-1 \end{bmatrix}.$$

   (a) Compute the characteristic polynomials of A and $P^{-1}AP$.
   (b) Show that all zeros of $p(t)$ are real regardless of the value of α.
9. Let A be a 4×4 non-invertible matrix with all diagonal entries 1. If $2 + 3i$ is an eigenvalue of A, what are its other eigenvalues?
10. Show that if each eigenvalue of $A \in \mathbb{F}^{n\times n}$ has absolute value less than 1, then both $I - A$ and $I + A$ are invertible.
11. Show that if rank of an $n \times n$ matrix is 1, then its trace is one of its eigenvalues. What are its other eigenvalues?
12. Find the spectrum of a matrix where the sum of entries in each row is same.
13. Let $A, B, P \in \mathbb{C}^{n\times n}$ be such that $B = P^{-1}AP$. Let λ be an eigenvalue of A. Show that a vector v is an eigenvector of B corresponding to the eigenvalue λ iff Pv is an eigenvector of A corresponding to the same eigenvalue λ.
14. Do equivalent matrices have same eigenvalues?
15. Let $A \in \mathbb{F}^{n\times n}$. Show that for all $x, y \in \mathbb{F}^{n\times 1}$, $\langle Ax, Ay\rangle = \langle x, y\rangle$ iff for all $x \in \mathbb{F}^{n\times 1}$, $\|Ax\| = \|x\|$.
16. Construct a 3×3 hermitian matrix with no zero entries whose eigenvalues are 1, 2 and 3.
17. Construct a 2×2 real skew symmetric matrix whose eigenvalues are purely imaginary.
18. Show that if an invertible matrix is real symmetric, then so is its inverse.
19. Show that if an invertible matrix is hermitian, then so is its inverse.
20. Construct an orthogonal 2×2 matrix whose determinant is 1.
21. Construct an orthogonal 2×2 matrix whose determinant is -1.

1. Suppose $A \in \mathbb{F}^{n\times n}$, $\lambda \in \mathbb{F}$, and $b \in \mathbb{F}^{n\times 1}$ are such that $(A - \lambda I)x = b$ has a unique solution. Show that λ is not an eigenvalue of A.

$Px = b$ has unique solution $\Rightarrow$ $\text{rank}(P|b) = \text{rank}(P) = n$

$\Rightarrow \text{rank}(A-\lambda I) = n \Rightarrow |A-\lambda I| \neq 0$

But if λ is eigenvalue it satisfies $|A-\lambda I| = 0$

$\Rightarrow \lambda$ isn't eigenvalue

4. Let $A \in \mathbb{F}^{n\times n}$. Show that A is invertible iff 0 is not an eigenvalue of A.

A is invertible $\Rightarrow Ax = \lambda x$

$\Rightarrow x = (A^{-1}\lambda)x$

$\Rightarrow \lambda$ can't be zero

0 is not an eigenvalue

$\Rightarrow Ax = \lambda x$

$(A-\lambda I)x = 0$

$\Rightarrow |A-\lambda I| = 0$

$\Rightarrow (\lambda-\lambda_1)(\lambda-\lambda_2)\dots(\lambda-\lambda_n) = |A-\lambda I| = 0$

$\Rightarrow \lambda_1\lambda_2\dots\lambda_n = |A|$

Since non of λ's are 0, $|A| \neq 0$

$\rightarrow$ A is invertible

6. Find all eigenvalues and corresponding eigenvectors of the following matrices:

(a) $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$

(a) $\begin{vmatrix} \lambda-3 & 0 & 0 & 0 \\ 0 & \lambda-2 & 0 & 0 \\ 0 & 0 & \lambda & 2 \\ 0 & 0 & -2 & \lambda \end{vmatrix} = 0 \Rightarrow (\lambda-3)\left[(\lambda-2)(\lambda^2+4)\right] = 0$

$\Rightarrow \lambda = 3, 2, 2i, -2i$

$\lambda=3$: $3x_1=3x_1$

$2x_1=3x_2 \quad x_2=0$

$-2x_4=3x_3 \quad x_3=0$

$2x_3=3x_4 \quad x_4=0$

$$\begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

$\lambda=2$: $\begin{bmatrix}0\\1\\0\\0\end{bmatrix}$ $\lambda=2i$: $\begin{bmatrix}0\\0\\i\\1\end{bmatrix}$ $\lambda=-2i$: $\begin{bmatrix}0\\0\\-i\\1\end{bmatrix}$

(b)
$$\begin{vmatrix} \lambda-1 & -2 & -3 & -4 & -5 \\ -1 & \lambda-2 & -3 & -4 & -5 \\ -1 & -2 & \lambda-3 & -4 & -5 \\ -1 & -2 & -3 & \lambda-4 & -5 \\ -1 & -2 & -3 & -4 & \lambda-5 \end{vmatrix}$$

$$=$$

$$\begin{vmatrix} \lambda-1 & -2 & -3 & -4 & -5 \\ -\lambda & \lambda & 0 & 0 & 0 \\ 0 & -\lambda & \lambda & 0 & 0 \\ 0 & 0 & -\lambda & \lambda & 0 \\ 0 & 0 & 0 & -\lambda & \lambda \end{vmatrix} = \begin{matrix}(\lambda-1)\lambda^4+2(-\lambda^4) \\ -3(\lambda^4)+4(-\lambda^4) \\ -5(\lambda^4)\end{matrix}$$

$$= \lambda^4[\lambda-1-2-3-4-5] = \lambda^4[\lambda-15]$$

$\Rightarrow \lambda=0, 15$

$\lambda=0$: $x_1+2x_2+3x_3+4x_4+5x_4=0$

$$\begin{bmatrix}-2\\1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}-3\\0\\1\\0\\0\end{bmatrix}, \begin{bmatrix}-4\\0\\0\\1\\0\end{bmatrix}, \begin{bmatrix}-5\\0\\0\\0\\1\end{bmatrix}$$

$\lambda = 15$: $15x_1 = 15x_2 = 15x_3 = 15x_4 = 15x_5 \Rightarrow \begin{bmatrix}1\\1\\1\\1\\1\end{bmatrix}$

7. Find all eigenvalues and their corresponding eigenvectors of the $n \times n$ matrix A whose jth row has each entry j.

$$A = \begin{bmatrix}1 & 1 & 1 & \cdots & 1\\ 2 & 2 & 2 & \cdots & 2\\ \vdots & & & & \\ n & n & n & \cdots & n\end{bmatrix}$$

c = each column of $A = \begin{bmatrix}1\\2\\\vdots\\n\end{bmatrix}$

$A = c \cdot [1 \ 1 \ 1 \ \cdots \ 1]$

$\lambda_1 = \frac{n(n+1)}{2} \Rightarrow v = [1 \ 1 \ 1 \ \cdots \ 1]^t$

$\lambda_2 = 0 \Rightarrow$ vector that is orthogonal to $[1 \ 1 \ \cdots \ 1]^t$

5. Let $A \in \mathbb{F}^{n \times n}$. If the entries in each column of A add up to a scalar λ, then find an eigenvector of A.

$$A^T \begin{bmatrix}1\\1\\\vdots\\1\end{bmatrix} = \lambda \begin{bmatrix}1\\1\\\vdots\\1\end{bmatrix} \Rightarrow \lambda \text{ is eigenvector of } A^T$$

10. Show that if each eigenvalue of $A \in \mathbb{F}^{n\times n}$ has absolute value less than 1, then both $I - A$ and $I + A$ are invertible.

$(I-A)v = v - Av = v - \lambda v = (1-\lambda)v$

for $I-A$ eigenvalue is $1-\lambda$

but $1-\lambda \in (0,2) \neq 0$ hence $I-A$ is invertible

same of $I+A$

11. Show that if rank of an $n \times n$ matrix is 1, then its trace is one of its eigenvalues. What are its other eigenvalues?

$\Rightarrow$ atmost 1 non-zero eigenvalue

$\Rightarrow \lambda, 0, \ldots (n-1), 0$

$tr(A) = \sum \lambda_i = \lambda$

3. Let $A \in \mathbb{F}^{n\times n}$. Show that a scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A iff the map $A - \lambda I : \mathbb{F}^{n\times 1} \to \mathbb{F}^{n\times 1}$ is not one-one.

If λ is eigenvalue: $(A-\lambda I)v = 0$

v_1 satisfies $\Rightarrow kv_1$ also does

$\Rightarrow$ not one-one

If F: isn't one-one:

kernel contains non-zero vector v

$\Rightarrow (A-\lambda I)v = 0$

Construct a 3×3 hermitian matrix with no zero entries whose eigenvalues are 1, 2 and 3.

$A = PDP^\dagger \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

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